Lab 3: Architecture Justification

Sanjana Aravamudhan | PES1UG24CS713 | PES University, CSE | Software Engineering

Architecture Selection

We chose Microservices Architecture for the Webhook Ingestion & Retry Mechanism Hub System.

Architectural Choice

The hub is split into small, independently deployable services (Ingestion Gateway, Durable Webhook Buffer, Delivery Service, Retry Scheduler, Dead-Letter Queue Manager, Logging & Audit Service, Endpoint Registry and Operator Console) that interact only through provided and required interfaces. A Layered design would keep ingestion, delivery and retries in one deployable unit, so slow targets and retries would compete with ingestion for the same resources. A Client-Server design would centralise everything on one server, a single point of failure that conflicts with the 99.999% buffering target (NFR-001).

Two Reasons

- **Independent scaling and fault isolation.** Ingestion must absorb up to 2,500 requests/second (NFR-002), while delivery speed depends on internal targets that may be slow or down. As separate services, gateway replicas scale for traffic bursts and delivery workers scale for backlog. A failing target only backs up the Delivery Service and Retry Scheduler; the gateway keeps accepting webhooks (FR-001).
- **Each requirement maps to one replaceable component.** Payload logging (FR-002), forwarding (FR-003), the 1s/2s/4s/8s backoff (FR-004) and dead-letter persistence (FR-005) each live in their own component behind an interface. The backoff policy or DLQ storage can change without redeploying the gateway. This matches the three Lab 2 epics: Webhook Ingestion and Processing, Delivery Retry and Reliability, and Dead-Letter Queue Management.

Security Advantage

Only the Ingestion Gateway is exposed to the outside world. It verifies the HMAC signature of every inbound webhook and rejects invalid requests before anything reaches the buffer. All other services sit on a private network and call each other over authenticated (mutual TLS) connections. Only the Delivery Service has outbound access to internal targets, and the DLQ Manager is reachable only through the role-restricted Operator Console. An attack or abuse at the edge therefore cannot reach the internal targets or the dead-letter data directly, which keeps the blast radius small and supports the security requirement in NFR-002.

Performance Benefit

The Durable Webhook Buffer decouples ingestion from delivery. The gateway acknowledges a webhook (202 Accepted) as soon as the payload is durably enqueued, so response time does not depend on how fast the target responds. Retries are scheduled as delayed re-enqueues instead of sleeping worker threads, so the 1s to 8s backoff never blocks capacity. This lets ingestion sustain 2,500 requests/second while deliveries and retries proceed at their own pace.

Trade-off Acknowledged

Microservices add operational complexity and network latency between services. We limit this by keeping the service count small (eight) and routing all asynchronous work through a single buffer.